

Table 1: Comparison of Lesion Analysis and Mapping Tools

Tool	Input	Output	Pros	Cons
VLSM	Lesion masks + behavioral scores	Lesion-symptom localization	Well-established; anatomy-behavior link	Needs large number of scans; sensitive to registration
LQT (MATLAB)	MRI lesion masks; atlas	Estimates region-level grey matter damage and white matter disconnection	Publicly available	Atlas accuracy
SVR-LSM	Lesion masks + behavior	ML model	Flexible; handles high-dimensional data	Needs behavioral data
DSM	Lesion masks + tract atlas	White matter disconnection maps	Highlights network disruptions	Requires diffusion atlas
CLSM	Lesion masks + connectome	Network lesion mapping	Network-level insight; connectomics	Needs high-quality connectome